

Computer Vision Test

Course :- Computer Vision

Code :- ITA0502

Name :- A. Lakshmi Devi

Reg No :- 192525176.

1) Sampling

The Problem is low spatial resolution, because the image is sampled with fewer pixels. Small objects become unclear. To improve it, increase the sampling rate / resolution by using a camera with more pixels or capturing the image at a higher resolution. This gives more detail and makes small objects clearer.

2) Dark and noisy image

Use image enhancement techniques such as brightness / contrast adjustment and ~~add~~ noise, while contrast enhancement makes dark objects more visible. This is suitable because it improves visibility while reducing unwanted noise.

3) Quantization

Quantization converts continuous intensity value into a fixed number of digital intensity levels. For ex, an image can use 256 levels from 0 to 255. This helps convert the continuous image into digital.

Values that a Computer Can store and process.

4) Uneven lighting

Use Contrast enhancement such as histogram equalization or adpotization histogram equalization. It improves the Contrast b/w the Product and its defects. This makes hidden defects more visible even when lighting is Uneven.

5) Image Segmentation

Image Segmentation divides the image into different regions such as roads, vehicles, and pedestrians. Each object can be identified separately. This helps a self-driving car understand its surrounding and make decisions such as stopping, turning or avoiding obstacles.

6) Edge detection

Apply an edge detection technique such as the can edge detector. It identifies sudden changes in intensity and produces the edges of objects. cracks appears as thin edge structures

making them easier than ~~the~~ to detect and ~~exam~~ inspect automatically.

7) filtering

Use a median filter to remove unwanted noise from the medical image. It reduces noise while preserving important edges. This makes the tumor and surrounding tissues clear and helps doctors or image processing systems identify the tumor more easily.

8) Object detection

Object detection can identify vehicles in each video frame and determine their position. The detected vehicles can be tracked across frames. This helps in traffic management by ~~monitoring~~ monitoring traffic flow, counting vehicles.

9) Motion blur

The possible cause is object movement during image acquisition, which produces motion blur. A suitable

technique is deblurring / restoration
Such as using a motion blur
model and inverse filtering.

10) Image classification.

Image classification can classify
Products into categories such
defective and good based on
their appearance. A trained
classification model learns the
visual features of both types.